

Wenbin Wang, Master.

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🌐 <http://WenbinWang12.github.io>

Research Interest

My research lies at the intersection of machine learning, statistics, and optimization, with a focus on the theoretical foundations of large language models and the development of AI algorithms for large-scale and multi-agent decision-making. I am particularly interested in understanding the mechanisms of large language models, including their training and prompting processes, and leveraging these insights for applications like decision-making. Recently, my work has focused on reinforcement learning from human feedback (RLHF) and the mathematical foundations of reinforcement learning.

Education

- 2023 – 2026  **M.Eng. Engineering, Computer Science and Technology** at ShanghaiTech University.
Advisor: Prof. Ziping Zhao
- 2018 – 2022  **B.Econ. Economics, Economic Statistics** at Huazhong Agricultural University.
Thesis title: *The Impact of Demographic Structure Change on the Economic Growth of China.*
Advisor: Prof. Zeyu Zhang

Research Publications

Journal Articles

- 1 **W. Wang** and Z. Zhao, “Compressive covariance matrix sensing via quadratic sampling based on nonconvex learning”, *Submitted to IEEE Trans. Inf. Theory*, 2025, Under review.
- 2 **W. Wang** and Z. Zhao, “Large low-rank covariance matrix estimation with oracle rate via majorization-minimization”, *Submitted to IEEE Trans. Signal Process.*, 2025, Under review.
- 3 **W. Wang***, Q. Ma*, Z. Zhang, Y. Liu, Z. Liu, and M. Fang, “Federated unlearning under attack: Poisoning threats and defensive strategies”, *Submitted to IEEE Trans. Inf. Forensics Security*, 2025, Under review.

Conference Proceedings

- 1 **W. Wang**, Z. Jing, Y. Shi, and Z. Zhao, “Low-rank covariance matrix estimation from quadratic measurements”, in *Proc. The Conference on Uncertainty in Artificial Intelligence (UAI)*, Under review, 2025.
- 2 **W. Wang**, Y. Shi, and Z. Zhao, “High-dimensional tensor regression with oracle properties”, in *Proc. The International Conference on Machine Learning (ICML)*, Under review, 2025.
- 3 **W. Wang**, J. Zhang, Y. Yan, J. Zeng, and Z. Zhao, “Achieving oracle statistical property for low-rank matrix recovery with application to image denoising”, in *Proc. The International Conference on Computer Vision (ICCV)*, Under review, 2025.
- 4 **W. Wang** and Z. Zhao, “Accelerating graph analytics via bilinear sketching”, in *Proc. Conf. Learn. Theory (COLT)*, 2025.
- 5 **W. Wang** and Z. Zhao, “Compressive covariance sketching: Efficient rank-1 sampling and optimal recovery”, in *Proc. The Int. Conf. Sampling Theory Appli. (SampTA)*, 2025. [\[Paper\]](#).
- 6 **W. Wang***, Q. Ma*, Z. Zhang, Y. Liu, Z. Liu, and M. Fang, “Poisoning attacks and defenses to federated unlearning”, in *Proc. The Web Conference (WWW)*, 2025. [\[Paper\]](#).

Skills

Languages	📖	Strong reading, writing and speaking competencies for English, Mandarin Chinese.
Coding	📖	C/C++, Python, R, $\LaTeX$, Tensorflow, Pytorch, MATLAB.
Developer Tools	📖	Git, Docker, Visual Studio Code, Jupyter Notebook, Linux Shell.
Misc.	📖	Academic research, teaching, training, $\LaTeX$ typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

2019	📖	Merit Award , Huazhong Agricultural University.
2020	📖	Practical Innovation Award , College of Economics and Management, Huazhong Agricultural University.
2022	📖	Outstanding Graduate , College of Economics and Management, Huazhong Agricultural University.

Teaching

Fall 2024	📖	Teaching Assistant . CS182 Introduction to Machine Learning.
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Professional Activity

2025	📖	Conference Reviewer . ICASSP2025.
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